📊 Titanic Dataset Analysis –Data Collection & Data Source Identification

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1. Project Overview

This task focuses on collecting and understanding the Titanic dataset, which is widely used for data analysis and machine learning. The dataset contains passenger information and is used to analyze survival patterns during the Titanic disaster.

2. Objective of Data Collection

The goal of this stage was to:

- Identify a reliable dataset
- Understand the structure and variables
- Prepare the dataset for cleaning and analysis

3. Data Source Identification

The dataset was obtained from:

Source: Kaggle Titanic Dataset

Type: Structured dataset (CSV file)

Format: Tabular data

4. Dataset Description

The dataset contains information about passengers aboard the Titanic.

Key Features (Columns):

- PassengerId – Unique ID for each passenger
- Survived – Survival status (0 = No, 1 = Yes)
- Pclass – Passenger class (1st, 2nd, 3rd)
- Name – Passenger name
- Sex – Gender
- Age – Age of passenger
- SibSp – Number of siblings/spouses aboard

- Parch – Number of parents/children aboard
- Ticket – Ticket number
- Fare – Ticket fare
- Cabin – Cabin number
- Embarked – Port of embarkation

5. Data Structure

Rows represent individual passengers

Columns represent features/attributes

Mixed data types:

- Numerical (Age, Fare)
- Categorical (Sex, Embarked)
- Text (Name, Ticket)

6. Initial Data Inspection

Before cleaning, the dataset was reviewed to:

- Understand missing values
- Check data types
- Identify inconsistencies

Observations:

Missing values in:

- Age
- Cabin
- Embarked
- Some columns (e.g., Cabin) have many null values
- Categorical data needs encoding

7. Data Collection Process

The following steps were followed:

Downloaded dataset from Kaggle

Loaded data into Power BI / Excel / Python

Verified dataset integrity

Prepared for cleaning using Python (Pandas, Numpy)

8. Data Quality Issues Identified

Missing values

Inconsistent formatting

Unnecessary columns (e.g., Cabin in some cases)

Mixed data types

9. Importance

This stage ensures:

- Reliable analysis
- Accurate insights

Smooth data cleaning process

Without proper data collection and understanding, analysis results can be misleading.

10. Conclusion

The Titanic dataset was successfully identified, collected, and understood.

This forms the foundation for the next stage: Data Cleaning & Transformation.

11. Relevance of Dataset

The Titanic dataset is ideal for:

- Learning data analysis
- Practicing data cleaning
- Understanding classification problems
- Building visuals